

PROBABILITY & COUNTING $\left(\frac{52}{5}\right) = 2598960$

Inclusion-Exclusion Principle: $|A \cup B| = |A| + |B| - |A \cap B|$

Permutations: # of permutations of k distinct elements from a set of size n . $n P_k = \frac{n!}{(n-k)!} = n(n-1)\dots(n-k+1)$

Combinations: # of combinations of k elements from a set of size n . $n C_k = \binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n P_k}{k!}$

Conditional Probability: $P(B) \neq 0$
 $P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$

DISCRETE RANDOM VARIABLE Probability Mass function (pmf): $p(a) = P(X=a)$

DISTRIBUTION	values, range	pmf	$E[X]$	$Var(X)$
Bernoulli(p)	0, 1	$p(0) = 1-p, p(1) = p$	p	$p(1-p)$
binomial(n, p)	0, 1, ..., n	$p(k) = \binom{n}{k} p^k (1-p)^{n-k}$	np	$np(1-p)$
uniform(a, b)	$a, \dots, b$	$p(k) = \frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
geometric(p)	0, 1, 2, ...	$p(k) = p(1-p)^k$	$\frac{1-p}{p}$	$\frac{1-p}{p^2}$

CONTINUOUS RANDOM VARIABLES

A random variable X is continuous if there is a function $f(x)$ s.t. for any $c \leq d$ we have $P(c \leq X \leq d) = \int_c^d f(x) dx$ $f(x)$: pdf.

$f(x)$: probability density function (pdf) similar to pmf, but note that it's not probability (have to integrate)
 * $f(x) \geq 0$ (f is nonnegative)
 * $\int_{-\infty}^{\infty} f(x) dx = 1 \iff P(-\infty < X < \infty) = 1$

$F(x)$: cumulative density function (cdf)
 $F(b) = P(X \leq b) = \int_{-\infty}^b f(x) dx$, where $f(x)$ is pdf of X .
 * $0 \leq F(x) \leq 1$ * $F'(x) = f(x)$
 * $F(x)$ is non-decreasing (if $a \leq b$, then $F(a) \leq F(b)$)
 * $P(a \leq X \leq b) = F(b) - F(a)$

PARAMETERIZED DISTRIBUTIONS

Uniform Distribution: $[a, b]$ Situations where all outcomes in range have equal probability
 pdf: $f(x) = \frac{1}{b-a}$ for $a \leq x \leq b$
 cdf: $F(x) = \frac{x-a}{b-a}$ for $a \leq x \leq b$

Exponential Distribution: $[0, \infty)$
 pdf: $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$ waiting time for a continuous process to change state
 cdf: $F(x) = 1 - e^{-\lambda x}$ for $x \geq 0$
 Tail Distribution: $P(X > x) = 1 - F(x) = e^{-\lambda x}$
 Normal Distribution: μ, σ $(-\infty, \infty)$

Normal (μ, σ^2)
 pdf: $f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$ intelligence, height, avg of lots of data
 cdf: $F(x) = \int_{-\infty}^x f(x) dx$
 Standard Normal $\downarrow$ $N(0, 1)$
 $Z = \frac{X-\mu}{\sigma}$

JOINT DISTRIBUTIONS

cdf:
 $F(x, y) = P(X \leq x, Y \leq y) = \int_c^y \int_a^x f(u, v) du dv$
 a, c : bottom ranges of X and Y respectively.
 $f(x, y) = \frac{\partial^2 F}{\partial x \partial y}(x, y)$ $[a, b] \times [c, d]$

Marginal pdf of X found by integrating out y over its full range: $f_X(x) = \int_c^d f(x, y) dy$
 Same for find marginal pdf of Y : $f_Y(y) = \int_a^b f(x, y) dx$
Marginal cdf: If $X+Y$ jointly $[a, b] \times [c, d]$ then $F_X(x) = F(x, d)$ $F_Y(y) = F(b, y)$

ex. pdf: $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$
 cdf: $F_X(x) = \int_0^x \lambda e^{-\lambda u} du = 1 - e^{-\lambda x}$
 $Y = X^2$ cdf: $F_Y(y) = \int_0^{\sqrt{y}} \lambda e^{-\lambda u} du = 1 - e^{-\lambda \sqrt{y}}$
 $X = \sqrt{Y}$ pdf: $f_Y(y) = \int_0^{\sqrt{y}} \lambda e^{-\lambda u} du = \frac{\lambda}{\sqrt{y}} e^{-\lambda \sqrt{y}}$
 ex. pmf of X if $X = \sum_{i=1}^n X_i$
 6-sided dice, \$3 if roll a 6
 pmf: $P(X=3k) = \binom{100}{k} \left(\frac{1}{6}\right)^k \left(\frac{5}{6}\right)^{100-k}$

Moving to Statistics:

Probability (Mathematics) $P(H|D) = \frac{P(D|H)P(H)}{P(D)}$
 Bayesian Path $\swarrow$
 $P_{\text{posterior}}(H|D) = \frac{P(D|H) P_{\text{prior}}(H)}{P(D)}$
 Statistics (Art) $\searrow$

Bayesians require a prior, so they develop one from the best information they have.

Independence: 2 events A & B independent if:

$P(A \cap B) = P(A)P(B)$
 Random variables $F(x, y) = F_X(x)F_Y(y)$
 Discrete variables $P(x_i, y_j) = P_X(x_i)P_Y(y_j)$
 Continuous variables $f(x, y) = f_X(x)f_Y(y)$ (factorable into x & y)
 Theorem: $X \& Y$ independent IFF $f_{X,Y}(x, y) = g(x)h(y)$
 Cumulative distribution function (cdf): $F(a) = P(X \leq a)$
 * F is non-decreasing. If $a \leq b$, then $F(a) \leq F(b)$
 * $0 \leq F(a) \leq 1$ $\lim_{a \rightarrow \infty} F(a) = 1, \lim_{a \rightarrow -\infty} F(a) = 0$

EXPECTATION & VARIANCE

$E[X] = \mu = \sum p(x_j) x_j$ $Var(X) = \sigma^2 = E[X^2] - E[X]^2 = E[X^2] - \mu^2$
 $E[ax+by] = aE[X] + bE[Y]$ $Var(X) = E[(X-\mu)^2] = \sum_{i=1}^n p(x_i) (x_i - \mu)^2$
 $E[X+Y] = E[X] + E[Y]$ $Var(aX+b) = a^2 Var(X)$
 $E[f(X)] = \sum p(x_j) f(x_j)$ $Var(X+Y) = Var(X) + Var(Y)$

Continuous Random Variables:

$E[X] = \int_a^b x f(x) dx$ $Var(X) = E[(X-\mu)^2] = \int_{-\infty}^{\infty} (x-\mu)^2 f(x) dx$

COVARIANCE

$Cov(X, Y) = E[(X-\mu_X)(Y-\mu_Y)]$
 positive covariance: when $X > \mu_X$, then Y tends to be $> \mu_Y$, vice versa
 ex. height & weight
 negative covariance: when $X > \mu_X$, then Y tends to be $< \mu_Y$, vice versa
 * $Cov(aX+b, cY+d) = ac Cov(X, Y)$
 * $Cov(X_1+X_2, Y) = Cov(X_1, Y) + Cov(X_2, Y)$
 * $Cov(X, X) = Var(X)$
 * $Cov(X, Y) = E[XY] - \mu_X \mu_Y$
 * $Var(X+Y) = Var(X) + Var(Y) + 2Cov(X, Y)$ for any X, Y
 * If $X \& Y$ independent $\Rightarrow Cov(X, Y) = 0$
 note: $cov(X, Y) = 0 \nRightarrow X, Y$ independent

CORRELATION

like covariance, but scale invariant
 $Cor(X, Y) = \rho = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$
 dimensionless (ratio)
 $-1 \leq \rho \leq 1$
 $\left\{ \begin{array}{l} \rho = +1 \iff Y = aX+b \text{ with } a > 0 \\ \rho = -1 \iff Y = aX+b \text{ with } a < 0 \\ \rho = 0 \rightarrow X \& Y \text{ independent (not the other way)} \end{array} \right.$
 correlation doesn't imply causation, vice versa.

Central Limit Theorem:

large # of independent, identically distributed random variables.
 $X_1, X_2, \dots, X_n$ $\mu = E[X_i]$ $\sigma^2 = Var(X_i)$
 As $n \rightarrow \infty$ $Z_n = \frac{\sum_{i=1}^n X_i - n\mu}{\sigma \sqrt{n}}$ approaches standard normal distribution $N(0, 1)$ (regardless of original distribution of X_i)

$\bar{X}_n = \frac{1}{n} \sum X_i \approx N(\mu, \frac{\sigma^2}{n})$ for large n .
 Rule of thumb: If you poll n , then margin of error is $\pm \frac{1}{\sqrt{n}}$
 $Z_n = \frac{S_n - n\mu}{\sigma \sqrt{n}} = \frac{\bar{X}_n - \mu}{\sigma / \sqrt{n}}$ $P(\sum_{i=1}^n X_i < y) = P(\frac{\sum X_i - n\mu}{\sqrt{n} \sigma} < \frac{y - n\mu}{\sqrt{n} \sigma})$

Continuous joint distribution ex:
 $f(x, y) = c(x+2y)$ $[0, 1] \times [0, 2]$
 $\int_0^1 \int_0^2 c(x+2y) dy dx = 1 \rightarrow c = 1/5$
 $F(a, b) = \int_0^a \int_0^b f(x, y) dy dx$
 $f_X(x) = \int_0^2 f(x, y) dy$ $f_Y(y) = \int_0^1 f(x, y) dx$
 independence $\Rightarrow f(x, y) = f_X(x) f_Y(y)$
 $E[X] = \int_0^1 \int_0^2 x f(x, y) dy dx$
 $E[Y] = \int_0^1 \int_0^2 y f(x, y) dy dx$
 $E[X^2+Y^2] = \int_0^1 \int_0^2 (x^2+y^2) f(x, y) dy dx$
 $Cov(X, Y) = E[XY] - E[X]E[Y]$
 $E[XY] = \int_0^1 \int_0^2 xy f(x, y) dy dx$ $\sigma_X = \sqrt{Var(X)}$
 $Cor(X, Y) = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$ $Var(X) = E[X^2] - E[X]^2$

Exponential ex:
 T : waiting time pdf: $f(t) = 2e^{-2t}$ on $[0, \infty)$
 pdf of the rate at which $R = \frac{1}{T}$:
 ① Find cdf: $F_T(t) = P(T \leq t) = \int_0^t 2e^{-2u} du = -e^{-2u} \Big|_0^t = 1 - e^{-2t}$ (of T)
 ② Find cdf: for $r > 0$: $(0, \infty)$ $F_R(r) = P(R \leq r) = P(\frac{1}{T} \leq r) = P(T > \frac{1}{r}) = 1 - F_T(\frac{1}{r}) = e^{-2/r}$ (of R)
 ③ Find pdf: $f_R(r) = \frac{d}{dr} (e^{-2/r}) = \frac{2}{r^2} e^{-2/r}$ (of R)

Memorylessness Ex: $f_X(x) = \lambda e^{-\lambda x}$
 ① $P(X \geq 5) = 1 - P(X \leq 5) = 1 - \int_0^5 \lambda e^{-\lambda x} dx = e^{-5\lambda}$
 ② $P(X \geq 15 | X \geq 10) = \frac{P(X \geq 15)}{P(X \geq 10)} = e^{-5\lambda}$ note: same.

Everyone uses Bayes' formula when the prior $P(H)$ is known.

Frequentist Path

Likelihood $L(H; D) = P(D|H)$

Frequentists draw inferences based only on the likelihood for observed and unobserved data.

Core Statistical Vocabulary:

Statistic: Any value computed strictly from the data (no unknown parameters)

- **Point Statistic:** A single value (e.g., sample mean $\bar{x}$, or MLE $\hat{\theta}$).
- **Interval Statistic:** A range (e.g. the span from min. to max. of $x_1, \dots, x_n$).
- **Set Statistic:** A set computed from data (e.g., the possible dice that could have rolled a 10).

Sampling Distribution: Because a statistic is computed from random data, the statistic itself is a random variable. Its probability distribution is called the sampling distribution.

Simple vs. Composite Hypotheses: Simple $\rightarrow$ Specifies the exact probability distribution (e.g., $H_0: \mu = 0$)
Composite $\rightarrow$ Specifies a range of probability distributions (e.g., $H_0: \mu > 0$)

Bayesian Inference:

Maximum Likelihood Estimate (MLE):

MLE finds the parameter θ that makes the observed data most probable.

1. **Likelihood function:** $L(\theta) = P(\text{Data}|\theta)$ Multiply probabilities (if independent).
2. **Log-likelihood:** $\ell(\theta) = \ln(L(\theta))$ Turns products to sums, drops exponents.
3. **Derivative:** Take $\frac{d}{d\theta} \ell(\theta)$
4. **Solve:** Set derivative to 0 and solve for $\theta \rightarrow \hat{\theta}$

Odds and Bayes Factor:

Odds: $O(E) = \frac{P(E)}{P(E^c)}$ (odds > 1 means more likely than not.)

Bayes Factor (BF) posterior odds

$BF = \frac{P(D|H)}{P(D|H^c)}$ Bayes $\times$ prior factor Odds

Generally 10 or bigger $\rightarrow$ Strong evidence

Conjugate Priors & Sketching:

Beta-Binomial: prior: $\beta(a, b)$ k successes, $n-k$ failures

posterior: $\beta(a+k, b+n-k)$

Normal-Normal: (known variance σ^2)

prior: $N(\mu_{\text{prior}}, \sigma_{\text{prior}}^2)$ Data: n samples, average $\bar{x}$

posterior mean:

$$\frac{\mu_{\text{post}}}{\sigma_{\text{post}}^2} = \frac{\mu_{\text{prior}}}{\sigma_{\text{prior}}^2} + \frac{n\bar{x}}{\sigma^2}$$

posterior variance:

$$\frac{1}{\sigma_{\text{post}}^2} = \frac{1}{\sigma_{\text{prior}}^2} + \frac{n}{\sigma^2}$$

$$\sigma_{\text{post}}^2 = \frac{1}{\frac{1}{\sigma_{\text{prior}}^2} + \frac{n}{\sigma^2}}$$

$$\sigma_{\text{post}}^2 = \frac{1}{\frac{1}{\sigma_{\text{prior}}^2} + \frac{n}{\sigma^2}}$$

Variance always decreases (more data) $\sigma_{\text{prior}}^2 > \sigma_{\text{post}}^2$

H	prior	likelihood	posterior
θ	$f(\theta) \sim N(\mu_{\text{prior}}, \sigma_{\text{prior}}^2)$ $= C_1 \exp(-\frac{(\theta - \mu_{\text{prior}})^2}{2\sigma_{\text{prior}}^2})$	$\phi(x \theta) \sim N(\theta, \sigma^2)$ $= C_2 \exp(-\frac{(x-\theta)^2}{2\sigma^2})$	$f(\theta x) \sim N(\mu_{\text{post}}, \sigma_{\text{post}}^2)$ $= C_3 \exp(-\frac{(\theta - \mu_{\text{post}})^2}{2\sigma_{\text{post}}^2})$

Sketching: Lower variance = skinnier, taller curve.

Z-test (For means, KNOWN population variance σ^2)

normal data: $x_1, \dots, x_n$

unknown mean μ , known variance σ^2

Hypotheses: $H_0: \mu = \mu_0$

H_A : Two-sided: $\mu \neq \mu_0$

Right-sided: $\mu > \mu_0$

Z-value: standardized $\bar{x}$: $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$

test statistic: z

Null distribution: Assuming $H_0: z \sim N(0, 1)$

Significance: α (often $\alpha = 0.05$)

Rejection Region:

Two-sided

Right-sided

Left and right tails (with prob $\alpha/2$ each)

Right tail, with prob α

P-value: two-sided $\rightarrow p = P(|Z| > |z| | H_0)$

right-sided $\rightarrow p = P(Z > z | H_0)$

Test: For $p \leq \alpha$, we reject H_0 in favor of H_A .

i.e. for $p \leq \alpha$, then z is in the rejection region.

Error, Significance, and Power:

Decision	True State	
	H_0 is true	H_A is true
Reject H_0	Type I Error	✓
Don't reject H_0	✓	Type II Error

Significance Level = $P(\text{test statistic in rejection region} | H_0)$

= $P(\text{type I error}) = P(\text{false positive}) = \alpha$

Power Level: $P(\text{test statistic in rejection region} | H_A)$

= $1 - P(\text{type II error}) = P(\text{true positive})$

Predictive Probabilities (Law of Total Probability):

Predicting the outcome of a future data point x :

• **Prior Predictive:** Probability of x before observing current data D : $P(x) = \sum P(x|\theta_i)P(\theta_i)$

• **Posterior Predictive:** Probability of x after updating beliefs with data D : $P(x|D) = \sum P(x|\theta_i)P(\theta_i|D)$

Bayesian Update Table:

	Hypothesis H	Prior $P(H)$	Likelihood $P(D H)$	Unnormalized posterior $P(D H)P(H)$	(normalized) posterior $P(H D)$
Discrete θ	θ	$p(\theta)$	$p(x \theta)$	$p(x \theta)p(\theta)$	$p(\theta x)$
Continuous θ	θ	$f(\theta)d\theta$	$p(x \theta)$	$p(x \theta)f(\theta)d\theta$	$f(\theta x)d\theta$
	total	1	-	Sum/integral of columns $P(D)$	1

A prior probability of 0 will always stay 0, no matter how much data you observe!

Credible Intervals & Quantiles:

• **Credible Interval:** A p-credible interval $[a, b]$ means there is a probability p that the true parameter θ falls within this range.

(Bayesian answer to confidence intervals) $P(a \leq \theta \leq b) = p$ (symmetric)

Discrete: pmf $p(\theta) \sum_{a \leq \theta \leq b} p(\theta) = p$ ex. 95% credible interval

Continuous: pdf $f(\theta) \int_a^b f(\theta) d\theta = p$ [$\mu_{\text{post}} - 1.96 \sigma_{\text{post}}, \mu_{\text{post}} + 1.96 \sigma_{\text{post}}$]

• **Constructing Subjective priors:** Estimate the median ($q_{0.5}$) and quantiles ($q_{0.25}, q_{0.75}$), then choose a prior distribution (like beta) that fits those boundaries.

• **Empirical Bayes:** When you have many parallel measurements, use the overall distribution to construct a common prior. This "regresses" individual estimates toward the global mean.

t-test (For means, UNKNOWN population variance σ^2)

one-sample t-test

$x_1, x_2, \dots, x_n \sim N(\mu, \sigma^2)$ $\mu + \sigma^2$ unknown

Null hypothesis: $\mu = \mu_0$ for some specific value μ_0

Test Statistic: $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$ where $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$

t : Studentized mean s^2 : sample variance

Null distribution: $\phi(t|H_0)$ pdf of $T \sim t(n-1)$

t distribution with $n-1$ degrees of freedom

One-sided p-value: Right Side $p = P(T \geq t | H_0)$

Left Side $p = P(T \leq t | H_0)$

Two-sided p-value: $p = \begin{cases} 2P(T \geq t) & \text{if } t > 0 \\ 2P(T \leq t) & \text{if } t < 0 \end{cases}$

can also express as: $p = P(|T| \geq |t|)$

(unpaired/independent)

two-sample t-test

Same variance!

Data: $x_1, x_2, \dots, x_n \sim N(\mu_1, \sigma^2)$

$y_1, y_2, \dots, y_m \sim N(\mu_2, \sigma^2)$

unequal variance

Null Hypothesis: $\mu_1 = \mu_2$

Test Statistic: $t = \frac{\bar{x} - \bar{y}}{s_p/\sqrt{\frac{1}{n} + \frac{1}{m}}}$

S_p^2 : pooled variance

$S_p^2 = \frac{(n-1)S_x^2 + (m-1)S_y^2}{n+m-2}$

S_x^2, S_y^2 : sample variances of x_i, y_i

Null distribution: $\phi(t|H_0)$ pdf of

$T \sim t(n+m-2)$

One-sided p-value:

Right side $p = P(T > t | H_0)$

Left side $p = P(T < t | H_0)$

Two-Sided p-value: $p = P(|T| > |t|)$

Assumes: drawn from normal distributions, equal variance

Paired two-sample t-test:

Data $x_1, x_2, \dots, x_n$ and $y_1, y_2, \dots, y_n$ (same length)

Differences $d_i = x_i - y_i$ are i.i.d. samples from $N(\mu_D, \sigma_D^2)$

Null Hypothesis: $\mu_D = \mu_0$ for some specific value μ_0 .

t-test: μ unknown, σ unknown

test statistic: (Studentized mean difference)

$t = \frac{\bar{d} - \mu_0}{s_d/\sqrt{n}}$, where $s_d^2 = \frac{1}{n-1} \sum_{i=1}^n (d_i - \bar{d})^2$

s_d^2 : sample variance for differences d_i .

Null distribution: $\phi(t|H_0)$ is the pdf of

$T \sim t(n-1)$

Confidence Intervals $P(\theta \in CI | \theta = \theta_0)$ = confidence

- Interval Statistic: Computed from sample data.
- Randomness: so confidence intervals are random
- Confidence level vs. Probability: confidence level (e.g. 95%) is a frequentist concept. Long term frequency that intervals generated by random trials will contain the true unknown parameter. It is never the probability that the unknown parameter is in a specific, already-computed interval.
- Significance Equivalence: Confidence level of an interval is analogous to the significance level (α) of a Null Hypothesis Significance Test (NHST), where the confidence level typically is $1-\alpha$.

Bootstrapping useful for when theoretical distribution hard to compute.

Setup & principle

- original data $x_1, \dots, x_n$ from unknown dist. F
- empirical dist. $F^* \approx F$ (by law of large numbers)
- Resample with replacement from data $\rightarrow$ bootstrap sample $x_1^*, \dots, x_n^*$
- variation of statistic $\mu^* \approx$ variation of μ .

Key idea: $S = \bar{x} - \mu$ (unknown)
Approximate with $S^* = \bar{x}^* - \bar{x}$ (computable)

Linear Regression

fitting a line: least squares

Model: $y_i = ax_i + b + \epsilon_i$

minimize $S = \sum (y_i - ax_i - b)^2$

$\hat{a} = \frac{\sum x_i y_i}{\sum x_i^2}, \hat{b} = \bar{y} - \hat{a}\bar{x}$

Where $S_{xx} = \sum (x_i - \bar{x})^2$

$S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y})$

Best fit line always passes through $(\bar{x}, \bar{y})$

Multivariate Linear Regression

Matrix formulation

Model: $y = X\beta + \epsilon, \epsilon \sim N(0, \sigma^2 I_n)$

X is $n \times (p+1)$ design matrix (first column all 1's for intercept)

$\beta = (\beta_0, \beta_1, \dots, \beta_p)^T, \beta = (X^T X)^{-1} X^T y$

Unique iff columns of X are linearly indepen. $\rightarrow$ least squares formula need $n \geq p+2$

2x2 Matrix Inverse

$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

Reading model.summary() output

Residual variance CI: $\left[\frac{(n-p-1)s^2}{C_{n/2}}, \frac{(n-p-1)s^2}{C_{1-n/2}} \right]$

using $\chi^2 (n-p-1)$

Correlation vs. Causation & Simpson's Paradox

Correlation $\neq$ Causation

- Regression coefficients = statistical relationships, not causal effects.
- Need randomized controlled trial (RCT) to establish causation.
- Confounders: variables correlated with both x and y can create spurious associations
- Classical Example: ice cream sales and drowning deaths both correlate with warm weather.

Simpson's Paradox

A trend appears in subgroups but reverses or disappears when groups are combined.

Berkeley admissions example: Overall, women appeared to be admitted at lower rates than men. But within each department, women were admitted at similar or higher rates. The paradox arose because women applied more to competitive departments.

Kidney Stones: Treatment A better for small stones AND large stones, but B looks better overall because A was used more on easy (small) cases.

Lesson: always check for confounders; can mislead.

CI Pivoting Tricks

z-CI for mean (known σ)

$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

μ_0 in CI $\Leftrightarrow$ don't reject H_0 at level α .

Bernoulli / Polling CI (non-normal data)

Conservative normal CI: $\bar{x} \pm \frac{z_{\alpha/2}}{2\sqrt{n}}$

Rule of thumb 95% CI: $\bar{x} \pm \frac{1}{\sqrt{n}}$

works because $\sigma = \sqrt{\theta(1-\theta)} \leq 1/2$ always.

Wider than necessary $\rightarrow$ conservative

Margin of error for polling n people $\approx \frac{1}{\sqrt{n}}$ (95%)

t-CI for mean (unknown σ)

$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$

$t_{\alpha/2}$ from $t(n-1)$. Use when σ unknown.

Connecting CI $\leftrightarrow$ NHST (pivoting)

Non-Rejection Region (centered on μ_0): $|\bar{x} - \mu_0| \leq \frac{z_{\alpha/2} \cdot \sigma}{\sqrt{n}}$

Pivot: flip roles of $\bar{x}$ and μ_0 .

Confidence Interval (centered on $\bar{x}$): $|\mu_0 - \bar{x}| \leq \frac{z_{\alpha/2} \cdot \sigma}{\sqrt{n}}$

μ_0 in CI $\Leftrightarrow \bar{x}$ in non-rejection region. Same width, different center.

Core Confidence Interval Formulas:

Interval Type	Estimating	Known Data	Formula
z-interval	Mean (μ)	Variance (σ^2)	$\left[\bar{x} - \frac{z_{\alpha/2} \cdot \sigma}{\sqrt{n}}, \bar{x} + \frac{z_{\alpha/2} \cdot \sigma}{\sqrt{n}} \right]$
t-interval	Mean (μ)	None (unknown σ^2)	$\left[\bar{x} - \frac{t_{\alpha/2} \cdot s}{\sqrt{n}}, \bar{x} + \frac{t_{\alpha/2} \cdot s}{\sqrt{n}} \right]$
χ^2 -interval	Variance (σ^2)	None (unknown σ^2)	$\left[\frac{(n-1)s^2}{C_{\alpha/2}}, \frac{(n-1)s^2}{C_{1-\alpha/2}} \right]$

Quantile: q_α (left tail) $P(X \leq q_\alpha) = \alpha$
Critical Value: C_α (right tail) $P(X \geq C_\alpha) = \alpha$

Conservative Normal CI: $\bar{x} \pm \frac{z_{\alpha/2}}{2\sqrt{n}}$
(works because of CLT and $\sigma = \sqrt{\theta(1-\theta)} \leq \frac{1}{2}$)

Empirical Percentile CI

- Generate B bootstrap means $\bar{x}_1^*, \dots, \bar{x}_B^*$
- sort them. Take $\alpha/2$ and $1-\alpha/2$ quantiles directly.
- $(1-\alpha)$ CI = $[q_{\alpha/2}(x^*), q_{1-\alpha/2}(x^*)]$

want same # of points as original to mimic the statistic

Nonlinear transforms $\rightarrow$ linear regression

If $y = ce^{ax}$, take $\ln$: $\ln(y) = ax + \ln(c)$

Regress $\ln(y)$ on x . Intercept $\hat{b} = \ln(c)$, so $c = e^{\hat{b}}$.

Goodness of fit: TSS, RSS, MSS, R^2

TSS = $\sum (y_i - \bar{y})^2$ (total variance in y)

RSS = $\sum (y_i - \hat{y}_i)^2$ (unexplained: residuals)

MSS = $\sum (\hat{y}_i - \bar{y})^2$ (explained by model)

TSS = MSS + RSS (pythagorean theorem)

$R^2 \in [0, 1]$. Closer to 1 = better fit. For Simple linear regression, $R^2 = (\text{corr coeff.})^2$

output field	what it means
estimate	$\hat{\beta}_i$ - least squares coeff.
std. error	$S\sqrt{V_{ii}}$; V_{ii} = i -th diagonal of $(X^T X)^{-1}$
t value	$\hat{\beta}_i$ / std. error testing $H_0: \beta_i = 0$
Pr(> t)	two-sided p-value for t-test on that coeff.
Residual std. err.	$S = \sqrt{RSS/(n-p-1)}$; estimates σ
Multiple R^2	$\text{Cor}(y, \hat{y})^2$
Adjusted R^2	$R^2 - \frac{(1-R^2)p}{n-p-1}$
F-Statistic	Tests $H_0: \beta_1 = \dots = \beta_p = 0$ simultaneously; avoids multiple comparison prob.

Examinable Study: Healthcare bias

What the algorithm did:

- Predicted future health care costs to identify high-risk patients for care management programs.
- Used by health systems nationwide; affected ~200M patients/year
- Race was explicitly excluded from features.

The Bias

- At any given risk score, Black patients had significantly more chronic illnesses than white patients.
- At 97th percentile: Black patients had 26.3% more chronic conditions than white patients at same score.
- Fixing bias would increase Black patients receiving extra care from 17.7% $\rightarrow$ 46.5%.

Examinable Study: UC Berkeley

Department was a confounder. Aggregating across departments hid the true pattern within each stratum.

Polling CLT: $R \sim N(p, \frac{p(1-p)}{n})$

Worst-case $\sigma_R = \frac{1}{2\sqrt{n}} > 95\%$ CI half width $\approx \frac{1}{\sqrt{n}}$

$\rightarrow$ need $n \geq \frac{1}{4\epsilon^2}$ Standardize: $Z = (R-p)/\sigma_R$

ex. $n = 400 \pm 5\%$
 $n = 10000 \pm 1\%$
 $n = 100 \pm 10\%$

Power of a z-test (μ known σ):

$P(\text{Reject } H_0 | H_a \text{ true}) = 1 - P(\text{type II})$

- Rejection Region under $H_0: \mu_0 \pm z_{\alpha/2} \cdot \sigma / \sqrt{n}$ (two-sided) $|\bar{x}| > \frac{z_{\alpha/2} \cdot \sigma}{\sqrt{n}}$
- Plug in alternative μ_A , compute $P(\bar{x} \text{ in rejection region} | \mu = \mu_A) = P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2} | \mu = \mu_A)$
- Standardize using μ_A and $\sigma / \sqrt{n}$: $Z = \frac{\bar{x} - \mu_A}{\sigma / \sqrt{n}}$
 $P(-z_{\alpha/2} - \frac{\mu_A - \mu_0}{\sigma / \sqrt{n}} \leq Z \leq z_{\alpha/2} - \frac{\mu_A - \mu_0}{\sigma / \sqrt{n}}) \approx 0.02$ Power = $1 - 0.02 = 0.98$

Empirical Basic CI

(Reverse Percentile)

- Compute $S^* = \bar{x}^* - \bar{x}$ for each bootstrap sample.
- Find $S_{\alpha/2}^*$ and $S_{1-\alpha/2}^*$ (quantiles of S^*)
- $(1-\alpha)$ CI = $[\bar{x} - S_{1-\alpha/2}^*, \bar{x} + S_{\alpha/2}^*]$ (pivoted, order flips!)

Homoscedasticity

Residuals ϵ_i have the same variance for all i .

Scatter plot: data hovers in band of fixed width around regression line. Required assumption for OLS.

Heteroscedasticity

Variance of residuals changes with x (e.g. spread fans out). $\rightarrow$ Need to transform data first.

Logistic Regression

model

for binary response $y \in \{0, 1\}$.

can't use linear regression (probabilities must be in $[0, 1]$).

$\text{logit}(p) = \ln \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$

$P(y=1|x) = \sigma(\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p) = \frac{e^{\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p}}{1 + e^{\beta_0 + \beta_1 x_1 + \dots + \beta_p x_p}}$

logit: maps $(0, 1) \rightarrow (-\infty, \infty)$

sigmoid σ : inverse, maps $(-\infty, \infty) \rightarrow (0, 1)$

fitted via MLE (not least squares)

using iterative numerical optimization (glm in R)

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Distribution/Critical Value

Standard Normal (Z): Right Critical Value $P(Z > z_{\alpha/2}) = \frac{\alpha}{2}$

Student t-distribution (t): $n-1$ degrees of freedom

Chi-Squared (χ^2): $n-1$ degrees of freedom

Rule of Thumb: 95% CI $\left[\bar{x} - \frac{2\sigma}{\sqrt{n}}, \bar{x} + \frac{2\sigma}{\sqrt{n}} \right]$

Margin of Error: $\frac{z_{\alpha/2} \cdot \sigma}{\sqrt{n}}$

n, c = confidence, $\bar{x}, \sigma$ width of a confidence interval:
 $\uparrow n$: narrower $\uparrow \mu$: stays the same
 $\uparrow c$: wider $\uparrow \sigma$: wider

Parametric Bootstrap CI

- If data is known to come from parametric family $F(\theta)$: fit $\hat{\theta}$ (e.g. MLE), then draw bootstrap samples from $F(\hat{\theta})$ (not from the data directly)
- Compute $S^* = \hat{\theta}^* - \hat{\theta}$ for each bootstrap sample; pivot to get CI for θ .
- Use when: you know the family. More accurate when empirical bootstrap correct.

Overfitting

Higher degree polynomial $\rightarrow$ always fits training data better ($R^2 \rightarrow 1$), but generalizes poorly to new data.

Degree $p = n-1$ polynomial passes through all n points exactly.

Balance: Model fit vs. complexity. Adjusted R^2 penalizes extra parameters: $\bar{R}^2 = R^2 - \frac{(1-R^2)p}{n-p-1}$ $\bar{R}^2$ decreases when adding uninformative features.

Regression to the mean:

In standardized coords, best fit line is $v = \rho u$, $|\rho| \leq 1$

Predicted y is always closer to $\bar{y}$ (relative to σ) than x is to $\bar{x}$.

Classification & Softmax

For K categories: multinomial logistic regression.

Softmax generalizes sigmoid: $\text{softmax}(t_1, \dots, t_K) = e^{t_i} / (e^{t_1} + \dots + e^{t_K})$

outputs a probability distribution over K classes.

key functions:

name	formula	range
logit	$\ln(p/(1-p))$	$(0, 1) \rightarrow \mathbb{R}$
sigmoid	$e^t / (1 + e^t)$	$\mathbb{R} \rightarrow (0, 1)$
softmax	$e^{t_i} / \sum e^{t_j}$	$\mathbb{R}^K \rightarrow \Delta^K$

Root Cause:

- Algorithm predicted costs, not illness - but these diverge by race.
- Black patients generate lower costs than white patients with the same health needs (~\$1801/year less), due to unequal access to care, structural barriers, systemic discrimination.
- So the algorithm "correctly" predicted costs but systematically underestimated illness severity for Black patients.

fix + lesson:

- Replacing cost label with health outcome label (# chronic conditions) reduced bias by 84%.
- Lesson: choice of proxy(label) is the most important decision in algorithm design. Correlation vs. causation: cost correlates with health overall, but fails as a proxy across racial groups due to structural inequality.
- Wrong idea that higher cost $\rightarrow$ more sick.

General Rule for Gambling:

$p > 0.5$: many small bets (minimal) since LLN

$p < 0.5$: house has edge, make few big bets (maximal)

$p > 0.5$: doesn't matter for expected value, but maximal has higher var, so high prob of hitting bigger target.